YINGHAO LIU (RUBY)

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EDUCATION BACKGROUND

COLUMBIA UNIVERSITY - *Master of Art in Statistics* - 09/2024-12/2025 **New York, NY, USA**
UNIVERSITY OF CALIFORNIA, BERKELEY (EXCHANGE) - *Economics* - 08/2022-12/2022 **Berkeley, CA, USA**
GUANGDONG UNIVERSITY OF FOREIGN STUDIES - *Finance(Investment)* - 09/2020-06/2024 **Guangzhou, Guangdong, China**

WORK EXPERIENCES

COLUMBIA UNIVERSITY, *Research Assistant* **01/2026-NOW**

- Supported experimental validation of Decoded Neurofeedback (DecNef) models by assisting with model testing, data preprocessing, and pipeline-level debugging.
- Participated in the design and implementation of a simplified brain signal simulator to mimic real-time neuroimaging data streams, facilitating controlled testing of neurofeedback algorithms.

SAFEROCK, *Data Science Intern* **05/2025-08/2025**

- Processed and standardized large-scale well-level oil production datasets, designing a robust preprocessing pipeline that improved data consistency and reduced missing-value errors for modeling workflows.
- Built an end-to-end analytics application in **Streamlit**, integrating feature engineering, model interpretation, and production-trend visualization, enabling engineers and executives to interactively evaluate wells performance and decline behaviors.
- Designed and compared forecasting models—including decline models, boosted tree model—to quantify accuracy differences across basins and well types, generating insights that informed production planning and resource allocation.
- Delivered model performance diagnostics and uncertainty assessments to stakeholders, establishing a repeatable evaluation framework for future predictive modeling initiatives.

UNITED NATIONS, *Financial Data Analyst Intern* **04/2025-05/2025**

- Rebuilt and reconciled departmental financial statements, enhancing reporting accuracy and improving month-end financial visibility for management.
- Managed over **100+ invoices, payments, and vendor schedules**, ensuring timely processing and compliance with UN financial controls.
- Assisted in developing **cash flow forecasts** and **integrated 3-statement financial models**, providing analytical support for budgeting and long-term financial planning.

HSBC GLOBAL SERVICE CENTER, *Financial Services Specialist Intern* **07/2024-08/2024**

- Conducted analytics-driven KYC reviews for **150+ customer profiles**, extracting risk features and identifying inconsistencies that informed compliance decisions.
- Built rule-based screening logic to flag high-risk behaviors, contributing to the escalation of **20+ suspicious transactions** and supporting a **25% reduction in fraud exposure**.
- Streamlined verification workflows by standardizing data checks across product lines, improving operational throughput and ensuring adherence to regulatory requirements.
- Partnered with risk officers to refine escalation thresholds, improving the accuracy and timeliness of fraud detection processes.

CCB PRINCIPAL ASSET MANAGEMENT CO., LTD. , *Data Analysis Intern* **08/2023-10/2023**

- Produced **10+ detailed reports and client presentations**, transforming complex financial and market information into actionable insights with a client satisfaction rate of **98%**.
- Monitored real-time market indicators, policy changes, and macroeconomic trends to support investment strategy discussions.
- Organized and visualized **200+ financial datasets**, ensuring high-quality inputs for portfolio analysis and investment decision-making.

PROJECTS

FRAUD DETECTION MODEL (MACHINE LEARNING) **03/2025**

- Built a credit card fraud detection system using logistic regression with threshold tuning to maximize Recall under imbalance; benchmarked **Random Forest** and **XGBoost**, achieving higher **F1** and more stable detection performance.

AIRBNB HOUSING RELATIONAL DATABASE (POSTGRESQL) **02/2025**

- Designed a scalable relational schema and engineered Python ingestion pipelines to ensure referential integrity; performed **SQL** benchmarking and analytics on pricing, occupancy, and host behavior to evaluate schema efficiency.

DETECTING POKÉMON TYPES (DEEP AND LIGHTWEIGHT CNNS) **04/2025**

- Explored the feasibility of classifying Pokémon types using image data, with potential applications in augmented reality, and visual merchandising. Implemented and evaluated two convolutional neural networks: **EfficientNetV2B0** and **MobileNetV2**.

RESEARCH

RESEARCH ASSISTANT - 08/2022-12/2022 **Berkeley, CA, USA**

Gathered qualitative and **quantitative data** on various aspects of the educational systems, including curriculum design, student performance metrics.Employed **statistical tools** to analyze the collected data, identifying patterns, correlations, and differences.

SKILLS

Programming Skills: Python (Pandas, NumPy, Scikit Learn), VBA, SQL (RDB, MySQL, MS SQL Server, PostgreSQL), R
Software & Computer Skills: Microsoft Office, Data Visualization (Matplotlib, Seaborn), Dashboards (Power BI, Qlik Sense)